

## Homework 2

This homework is due on **Monday, Oct. 14, 2024 by 9:00pm**. Complete this assignment in your own notebook and then upload it as a single, fully executed PDF to Gradescope > Homework 2. Email submissions and/or submissions in any format other than PDF **will not be accepted**. You must clearly identify which question you are answering throughout your homework. If we cannot tell what question you are answering, we cannot give you credit.

This homework is worth 55 points.

Note that the course academic honesty policy applies to **every** assignment, including this one.

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### 1. Births in the US

For this question, we will use the dataset located at

`shared/data/US-births_2000-2014.csv`

This dataset contains the number of births that occurred in the United States on every day between January 1st, 2000 and December 31st, 2014. Note: if you were born in the U.S. between those dates, you are accounted for in this dataset!

Throughout this problem, we wish to investigate whether there are any time-based trends in births or if the people born during this time (62,187,024) are born purely at random across the 14 years.

Please load the dataset as a dataframe. Examine the dataset and its columns to acquaint yourself. Please set `parse_dates = ['timestamp']` in order to treat the `timestamp` column as a pandas `datetime` object when you call `pandas.read_csv()`.

- (a) **(2 points)** First, let's establish what we can expect if births occur at the same rate year-round (i.e., uniformly at random). If birthdays are equally likely, what fraction of them should occur on any given day of the week? Approximately what fraction should occur in any given month?
- (b) **(3 points)** Please find the number of births that occurred on each **day of the week** and divide it by the total number of births to find the proportion of births per day of the week. Draw a red horizontal line to show what fraction of births should occur under a uniform birth rate.

Do the empirical proportions match the theoretical line? Please provide an explanation why or why not this is the case and explain your reasoning.

- (c) **(3 points)** Please find the number of births that occurred **in each month** and divide it by the total number of births to find the proportion of births per month. Draw a red horizontal line to show what fraction of births should occur under a uniform birth rate. Use log-scaling on the  $y$ -axis this time
- Do the empirical proportions match the theoretical line? Please provide an explanation why or why not this is the case and explain your reasoning.
- (d) **(2 points)** Create a **scatterplot** of the `births` column on the  $y$ -axis against the `timestamp` column on the  $x$ -axis. Set `s=2` so there is less overlap between dots. Compute the expected number of births on each day under a uniform birth-rate by dividing the total number of births by the length of time for which the data was collected and plot it using a red horizontal line.
- (e) **(2 points)** There are two groups of points spanning the entire time period, one with a markedly lower amount of births than the other. What do you think the lower group corresponds to? Create another scatterplot that shows the bottom group in a different color than the top group to confirm your hypothesis.
- (f) **(2 points)** Comment on the birth rate as a function of time. What relevant events might explain what you observe?
2. In this question you will work with data from the Framingham Heart Study, which looks at risk factors on cardiovascular disease on subjects from Framingham, MA.
- (a) **(1 point)** Use `pandas` to load the dataset from `shared/data/framingham.csv`. Display the first 5 rows of the data.
- (b) **(2 points)** We'll be working specifically with the `cigsPerDay` column of this data. Plot the distribution of `cigsPerDay` as a histogram and comment on its skew.
- (c) **(2 points)** Focusing only on subjects who are current smokers, plot the distribution of `cigsPerDay` as a histogram. Comment on the differences between this plot and the previous plot.
- (d) **(2 points)** Using the full dataset, report the following summary statistics for the `cigsPerDay` column, rounding to 1 decimal place if needed.
- Number of observations
  - Mean
  - Median
  - Maximum
  - Standard deviation

- (e) **(2 points)** Suppose you introduce a typo in the `cigsPerDay` data; instead of 30 cigarettes, you accidentally change the value to 300. Describe how this typo would impact the mean compared to the median. How would the standard deviation change?
- (f) **(2 points)** Randomly sample 100 observations of `cigsPerDay`. Report the mean of the **sampled** data (i.e. the sample mean), rounding to 2 decimal places if needed.
- (g) **(2 points)** Take a sample of 100 observations of `cigsPerDay` **50 times**. For each sample of 100, calculate and save the mean. You should have 50 sample means. Plot the distribution of sample means as a histogram. Include in your plot (i) a vertical line indicating the true mean, (ii) a vertical line indicating the mean of the sampled means, and (iii) a legend noting which line is which. Remember to title your plot and label your axes.
- (h) **(2 points)** Take a sample of 100 observations of `cigsPerDay` **1,000 times**. Reproduce the steps in 5(e) and generate a new histogram showing the distribution of sample means from your 1,000 samples, remembering to title your plot and label your axes.
- (i) **(1 point)** What do you notice about the mean of the sampling distribution as the number of samples increases?
- (j) **(1 point)** What do you observe about the shape of the sampling distribution as the number of samples increases?
- (k) **(2 points)** What principle predicts the outcomes you see regarding both the mean and shape of the sampling distribution as the number of samples increases?

3. **Federalist Papers revisited** In this question, we will examine more closely the Federalist Papers dataset using the dataset of *all word counts* located at

`shared/data/federalist-papers/all_word_counts.csv`

- (a) **(1 point)** Please read the CSV file into your Jupyter notebook. You may wish to display the dataframe for your own inspection.

The last 5 columns: `Essay`, `Author`, `Publication`, `Date`, and `total_word_count` are metadata for each essay.

The rest of the columns are the word counts. Please print the first and last 5 of the counted words.

- (b) **(4 points)** Next we will work towards generating a boxplot of the `total_word_counts` for each essay, stratified by the `Author` using `matplotlib.pyplot.boxplot`.

To do so, you will need two lists:

- A list of unique `Authors`.
- A list of lists of integers such that its  $i$ -th item is the list of `total_word_counts` for the  $i$ -th `Author` of the first collection described in the previous bullet.

You may construct the two lists above however you choose, for example using `pandas.groupby`. Whatever your solution is, you must explain in your own words how it was done.

- (c) **(4 points)** With the grouped total word counts per authors, plot the spreads using `matplotlib.pyplot.boxplot`. Provide the unique author list as the `labels` argument and set `vert = False`.

Remember to label the  $x$ -axis and set a title to the plot. State two features of the information you plotted that you find interesting.

- (d) **(2 points)** Let's try to generate our own Federalist paper. Follow the instructions below and answer the following question:

Is the result a "realistic" essay? Why or why not do you think this is the case?

- i. Filter the dataframe so that only rows for your favorite author remain.
- ii. Isolate only the word columns (i.e., all but the last 5 columns) and sum over the rows so that you have the total number of occurrences of each word out of all the essays written the selected author.
- iii. Find the grand total of all the word counts and divide the individual total word counts (the result of (ii)) by the grand total. The result is *relative frequencies* of each word.
- iv. Using `np.random.choice()`, sample a new collection of 250 words using the relative frequencies from (iii) (you provide a `p` argument here).
- v. Use `" ".join` on the resulting array from (iv) and print it.

4. Consider the following lottery. Entry tickets cost \$4 and 100 individuals (including you) each buy one single ticket. The lottery draws (or samples) three tickets with no preference towards any contestant and tickets drawn win \$100. Make sure to explain your answer for each problem.
- (a) **(2 points)** What does it mean if the lottery draws (or samples) the winners with replacement? What is the probability the lottery outcome results in total winnings of \$300 to one individual?
  - (b) **(1 point)** What's the probability that an individual comes out of the lottery a winner (i.e. wins at least once)?
  - (c) **(2 points)** Assume that the lottery draws (or samples) its three winners *without* replacement for the rest of Problem 1. How is this scenario different than the previous one? What is the probability that you win 100 after the three draws?
  - (d) **(2 points)** You enter the lottery with two of your friends. What is the probability that all three of you win, assuming sampling without replacement?
  - (e) **(2 points)** Compare the expected winnings per ticket under both the 'with replacement' and 'without replacement' scenarios. Are they different? Explain why or why not.
  - (f) **(2 points)** This lottery happens once per day, and you play every day for a long time. What does the law of large numbers tell you about your average profit (winnings – cost of entry) per entry?